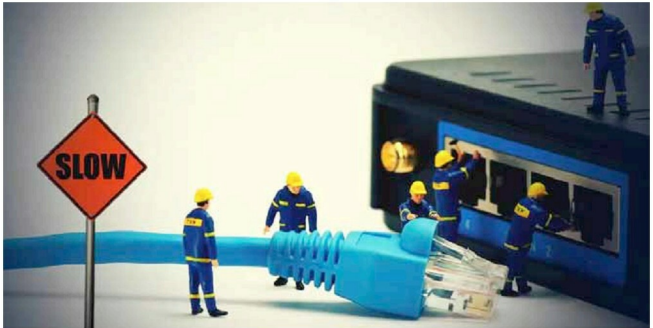


Is Your Network Getting Slower?

This article addresses a very vexing problem that plagues most of us at some time or the other. Internet speeds are a bone of contention between the user and the ISP. The author gives examples of how problems linked to a slow network were resolved using Wireshark.



Hopefully, readers have learnt from and enjoyed the earlier articles in this series on Wireshark, which started in July 2014. The articles included discussions on various important and interesting Wireshark features and various protocols such as ARP and DHCP, capturing packets in a switched environment, detecting an ARP spoofing attack, etc. This article will discuss how to put Wireshark to use when troubleshooting 'slow network' problems.

Computer users often say, "The network is slow" or "The Internet is working very slowly." Field engineers trying to troubleshoot such problems face an uphill task; first, to figure out what exactly a 'slow network' means and further, to locate and solve the actual problem. There are just too many different aspects and variables to check.

The actual reasons for slow network speeds could be broadly classified as follows.

Network related issues:

- Slow Internet bandwidth available from the ISP, especially in shared connections
- A faulty Ethernet switch
- A fault in the Ethernet interface on the switch
- A defective Ethernet card on a computer system or device in the network
- The presence of electrical noise in the vicinity of the network's cabling

Problems linked to the operating system, drivers and everything else:

- Ethernet device driver
- Presence of worms (a type of malware which generates a lot of network traffic)
- Insufficiently configured computer hardware
- Too many downloads initiated from a single system or network
- All systems scheduled to update the anti-virus or the operating system simultaneously
- Simultaneous use of ERP software during a particular time

To arrive at the correct troubleshooting approaches, consider the fundamental basis of networking — the OSI/TCP-IP layers. I assume that readers are knowledgeable about them - a lot of excellent information is also available on the Internet. Still, let me describe these layers in brief for relevance and immediate reference.

OSI layers separate network communication into seven different layers, each offering its services to its immediate upper and lower layer. This enables structured and standardised communication between systems. The OSI model does not require presence of each one of these seven layers. Also, multiple adjacent layers could be combined to form one layer. As of today, the most widely used model is TCP/IP, which has four layers. Do go through Table 1, which describes the four TCP/IP layers, their functions and corresponding OSI layer(s).

These OSI and TCP-IP layers are important from troubleshooting perspective—depending upon the situation, one may use these layers as guidelines when selecting a particular approach while troubleshooting.